Assignment - 1

Analysis of One Published Biological Dataset Using the Data Science Workflow

GTEx V8 RNA-seq Gene Expression Dataset

Dataset selected: The Genotype-Tissue Expression (GTEx) V8 dataset, a published large-scale human transcriptomic resource containing RNA-seq measurements from multiple postmortem tissues. It is appropriate for this assignment because it demonstrates the complete data science workflow: defining a biological question, obtaining biological data, preparing and cleaning the data, performing exploratory analysis, applying analytical methods, visualizing results, and interpreting the findings biologically.

The GTEx V8 release contains approximately 17,382 RNA-seq samples from 948 postmortem donors across 54 tissue sites and two cell lines. The final GTEx Consortium publication in Science (2020) reports analyses of a quality-controlled subset of the V8 data; for its QTL analyses, the study used 15,201 samples from 838 donors across 49 tissues or cell lines for which the required genotype and RNA-seq data were available.

1. Research Question

Main research question: How does gene expression differ among human tissues, and can RNA-seq expression profiles be used to identify tissue-specific biological patterns?

Specific questions include:

- Which genes show high or low expression in different tissues?
- Do samples from the same tissue have similar gene-expression profiles?
- Can exploratory methods such as Principal Component Analysis (PCA) and clustering separate samples according to tissue type?
- Which genes contribute most strongly to differences between tissues?

2. Data Source

The dataset is the Genotype-Tissue Expression (GTEx) Project V8 release. GTEx is an NIH Common Fund project created to characterize gene expression and genetic regulation across human tissues.

The V8 public release contains approximately 17,382 RNA-seq samples from 948 donors, representing 54 tissue sites and two cell lines. The data and associated resources are available through the GTEx Portal, with protected genotype and raw sequencing resources available through controlled-access repositories such as dbGaP.

Published dataset paper: The GTEx Consortium. (2020). The GTEx Consortium atlas of genetic regulatory effects across human tissues. Science, 369(6509), 1318–1330. doi:10.1126/science.aaz1776.

3. Data Type

The primary data type is bulk RNA-seq gene-expression data. The expression data can be represented as a gene × sample matrix, where rows represent genes, columns represent biological samples, and values represent gene-expression measurements.

The GTEx resources include gene-level expression measurements such as read counts and normalized expression values such as TPM. The dataset is therefore a high-dimensional quantitative biological dataset. In addition to expression measurements, GTEx contains sample and donor metadata and, in the broader project, genotype information.

4. Data Collection Method

GTEx biospecimens were collected from postmortem human donors. Multiple tissue sites were sampled, and RNA was isolated from tissue samples and subjected to RNA sequencing.

The general biological and computational workflow is:

Human tissue → RNA extraction → RNA sequencing → sequencing reads → alignment and quantification → quality control → gene-expression measurements

In the published V8 study, tissue RNA was sequenced to a median depth of approximately 82.6 million reads per sample. The resulting sequencing data were aligned, quantified, and quality-controlled before downstream analyses.

5. Data Cleaning Steps

Data preparation is essential because RNA-seq datasets can contain low-quality samples, lowly expressed genes, missing metadata, outliers, and technical variation. The following cleaning and preparation steps would be applied:

1. Check sample metadata for consistent sample identifiers, tissue labels, donor identifiers, sex, and other relevant variables.
2. Identify and remove or exclude samples that fail quality-control criteria.
3. Filter genes with extremely low expression because they provide limited information for many downstream analyses.
4. Identify missing values in expression data and metadata and handle them according to their biological and statistical context.
5. Normalize expression measurements when comparing samples so that differences in sequencing depth or library size do not create misleading comparisons.
6. Transform expression values when appropriate, for example using log2(expression + 1), for visualization and some exploratory analyses.
7. Detect potential outlier samples using sample correlations, PCA, and hierarchical clustering.
8. Assess technical variables and potential batch effects before interpreting biological differences.

6. Exploratory Analysis

Exploratory Data Analysis (EDA) is performed before formal modeling to understand the structure, distribution, quality, and major sources of variation in the dataset.

- Summary statistics: calculate the number of genes and samples, measures of central tendency, dispersion, and the number of detected genes per sample.
- Expression distributions: use histograms or density plots to inspect the distribution of expression values.
- Boxplots: compare expression distributions across samples or tissues and identify unusual samples.
- Sample correlation: calculate pairwise correlations to assess whether samples from the same tissue show similar expression profiles.
- Principal Component Analysis (PCA): reduce thousands of expression variables to a smaller number of dimensions and visualize major sources of variation.
- Hierarchical clustering: group samples according to similarities in their gene-expression profiles.

A successful exploratory analysis would be expected to show strong tissue-associated structure, with samples from biologically similar tissues tending to occupy nearby regions in PCA or clustering space.

7. Analysis or Modeling Method

The primary analytical methods selected for this assignment are Principal Component Analysis (PCA) and hierarchical clustering. These methods are appropriate because the research question focuses on discovering patterns in a high-dimensional gene-expression dataset rather than predicting a predefined clinical outcome.

7.1 Principal Component Analysis (PCA)

PCA transforms the original high-dimensional expression matrix into a smaller set of principal components. PC1 and PC2 can be plotted to determine whether samples separate according to tissue type or other major sources of variation.

7.2 Hierarchical Clustering

Hierarchical clustering groups samples based on expression similarity. A dendrogram can be used to examine whether samples from the same tissue cluster together and whether biologically related tissues show similar expression patterns.

A supervised classifier such as Random Forest could be added as an extension to predict tissue type, but it is not necessary to answer the primary research question. Keeping PCA and clustering as the main methods makes the analysis focused on exploratory biological pattern discovery.

8. Visualization Approach

The following visualizations would communicate the analysis clearly:

- Boxplots : compare expression distributions among tissues or samples.
- PCA scatter plot : show sample separation using PC1 and PC2.

- Heatmap : display expression levels of highly variable or tissue-specific genes across selected samples.
- Hierarchical clustering dendrogram : display relationships among samples based on expression similarity.
- Bar plot : summarize sample numbers by tissue or selected gene-level summaries.
- Volcano plot : if a specific pairwise tissue comparison is performed, display the relationship between expression change and statistical significance.

9. Biological Interpretation

The central biological expectation is that human tissues have distinct transcriptional programs. Genes required for the specialized functions of a tissue are expected to be more highly expressed in that tissue than in many other tissues.

- Muscle tissue is expected to show expression of genes involved in muscle structure and contraction.
- Brain tissues are expected to show expression of genes involved in neuronal function, signaling, and synaptic processes.
- Liver is expected to show expression of genes associated with metabolism and detoxification.
- Blood and immune-related tissues are expected to show expression of genes involved in immune and hematopoietic functions.

If PCA and clustering separate samples according to tissue, the computational result supports the biological conclusion that tissue identity is a major source of variation in gene expression. The analysis therefore connects computational patterns to biological function: expression differences → tissue-specific genes → biological pathways → tissue specialization.

10. Limitations and Challenges

Postmortem sampling: GTEx tissues are obtained after death, so some expression patterns may differ from those in living individuals.

Tissue heterogeneity: Bulk RNA-seq measures an average signal across the different cell types present in a tissue and may therefore hide cell-type-specific patterns.

Biological variability: Individuals differ genetically and physiologically, which contributes to variation in gene expression.

Technical variation: Differences in RNA quality, sample processing, sequencing, and other technical factors can introduce unwanted variation and batch effects.

High dimensionality: Thousands of genes are measured simultaneously, creating a high-dimensional analysis problem and increasing computational requirements.

Unequal tissue sample sizes: Different tissues do not necessarily have identical numbers of samples, which can affect statistical power and comparisons.

Confounding factors: Age, sex, ancestry, tissue composition, and other donor characteristics can contribute to observed expression differences.

Interpretation of association: An observed expression difference or association does not by itself establish a causal biological mechanism.

Generalizability: Because GTEx is primarily based on non-diseased postmortem tissues, findings cannot automatically be generalized to all disease states or living populations.

Overall Data Science Workflow

9. Define the biological question.
10. Identify the GTEx V8 data source and relevant metadata.
11. Obtain and organize the RNA-seq expression data.
12. Perform quality control, filtering, normalization, and data preparation.
13. Conduct exploratory data analysis using summary statistics, distributions, correlations, PCA, and clustering.
14. Analyze major expression patterns using PCA and hierarchical clustering.
15. Evaluate whether observed sample groupings are biologically plausible and consistent with tissue labels.
16. Visualize the findings using PCA plots, heatmaps, boxplots, dendrograms, and other appropriate figures.
17. Interpret the observed patterns in terms of tissue-specific biology.
18. Communicate the findings, limitations, and conclusions.

Conclusion

The GTEx V8 RNA-seq dataset provides an appropriate published biological dataset for demonstrating the data science workflow in bioinformatics. It contains large-scale gene-expression measurements from multiple human tissues and therefore illustrates the challenges and opportunities associated with high-dimensional biological data.

By defining a clear biological question, preparing the data, performing exploratory analysis, applying PCA and clustering, visualizing patterns, and interpreting the results biologically, the workflow transforms raw or processed RNA-seq measurements into meaningful biological knowledge. The analysis also demonstrates why data cleaning, reproducibility, appropriate visualization, and careful interpretation are essential in bioinformatics.

References

- The GTEx Consortium. (2020). The GTEx Consortium atlas of genetic regulatory effects across human tissues. Science, 369(6509), 1318–1330. https://doi.org/10.1126/science.aaz1776
- National Institutes of Health (NIH) Common Fund. Genotype-Tissue Expression (GTEx) Project. https://commonfund.nih.gov/GTEx
- GTEx Portal. GTEx V8 release and data resources. https://gtexportal.org/
- Introduction to Data Science, Module 1: Data Science and Cloud Computing in Bioinformatics. Course lecture material.